

Time Series Analysis of the Chicken Dataset

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I. Preliminary Discussion and Analysis

This report analyzes the chicken time series from the `astsa` package to identify an appropriate ARIMA model for forecasting the monthly U.S. chicken price index. The analysis includes exploratory data analysis, stationarity assessment, model identification, parameter estimation, model diagnostics, comparison with a competing model, and forecasting with prediction intervals. The dataset consists of 180 monthly observations from August 2001 to July 2016 and has a seasonal period of 12.

1. Exploratory Analysis of the Original Series

The time plot provides an initial assessment of the trend, variability, possible seasonality, and potential outliers before formal statistical modeling begins.

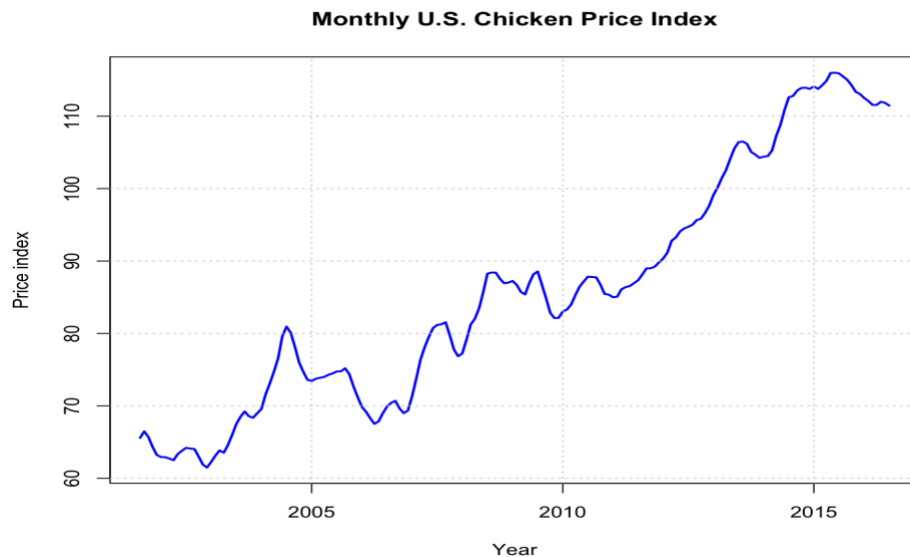


Figure 1. Monthly U.S. chicken price index from August 2001 to July 2016

The original series exhibits a clear long-term upward trend, increasing from approximately 65 to above 110 by the end of the sample. Although the series contains several periods of increases and declines, there is no clear evidence of a strong or consistent annual seasonal pattern. The changing level indicates non-stationarity in the mean. Since the variability does not appear to increase systematically with the level of the series, a Logarithmic or Cox-Box transformation is not necessary. In addition, there is no obvious outliers in the original series.

The next step is to investigate the presence of seasonal behavior using a month plot and the Augmented Dickey–Fuller test before determining the appropriate differencing and model order.

2. Examination of Seasonality

Figure 2 displays the month plot of the original chicken series.

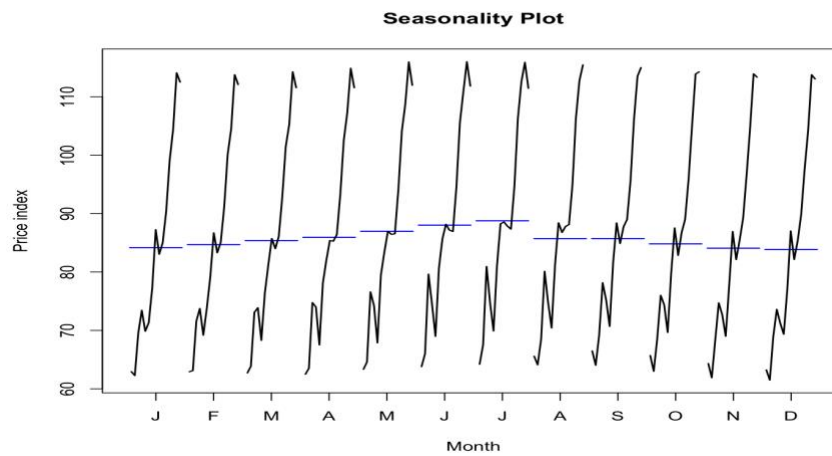


Figure 2. Month plot of the monthly U.S. chicken price index from August 2001 to July 2016

The monthly averages vary only slightly across the year, indicating a weak seasonality compared with the strong upward trend observed in the original series. Therefore, monthly frequency alone doesn't justify seasonal differencing. The initial analysis proceeds with one ordinary difference ($d = 1$) and no seasonal difference ($D = 0$), while allowing seasonal AR or MA terms capture any remaining dependence at annual lags. The absence of seasonal differencing does not imply the absence of seasonal correlation.

3. First Differencing and Stationarity

Figure 3 displays the first-differenced chicken series, where $\nabla X_t = X_t - X_{t-1}$

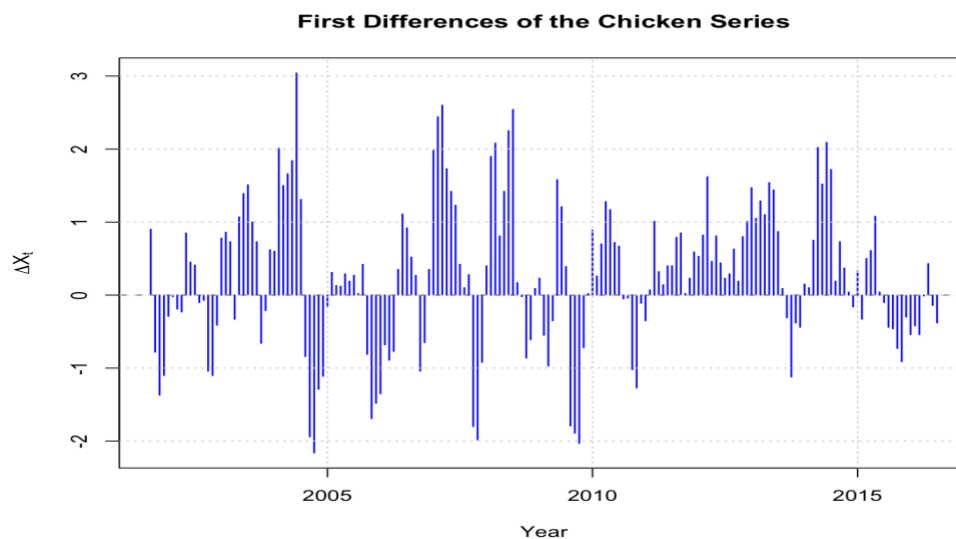


Figure 3. First Differences of the monthly U.S. chicken price index

The first-differenced series fluctuates around a stable level with a constant variance. The upward trend has been removed and its sample mean of the first differences is around 0.256. There is no visual evidence that a second ordinary difference or a seasonal difference is needed. The Augmented Dickey–Fuller test in the next section provides formal statistical confirmation of this conclusion.

4. Augmented Dickey-Fuller Test

Augmented Dickey–Fuller (ADF) test with 5 lags were applied to the original and first-differenced series. Five lags were included in the ADF tests based on the rule of thumb $(180 - 1)^{1/3} = 5.64$

	ADF statistic	p-value	Conclusion
Original series	-0.4599	0.8852	Fail to reject unit root
First differences	-5.6035	Less than 0.01	Reject unit root

For the original series, the ADF statistic is -0.4599 with a p-value of 0.8852 . So, we fail to reject the null hypothesis of a unit root (non-stationary) which is consistent with the upward trend observed in Figure 1. For the first-differenced series, the ADF statistic is -5.6035 with a p-value below 0.01 . The unit-root null hypothesis is rejected, providing evidence that the first differences are stationary.

The graphical analysis and the ADF results support the use of one ordinary difference ($d = 1$) and since the first-differenced series appears stationary without applying $(1 - B^{12})$, no seasonal difference is used $D = 0$.

5. Sample ACF and PACF of the Differenced Series

After applying one ordinary difference, the sample ACF and PACF were examined to identify model orders.

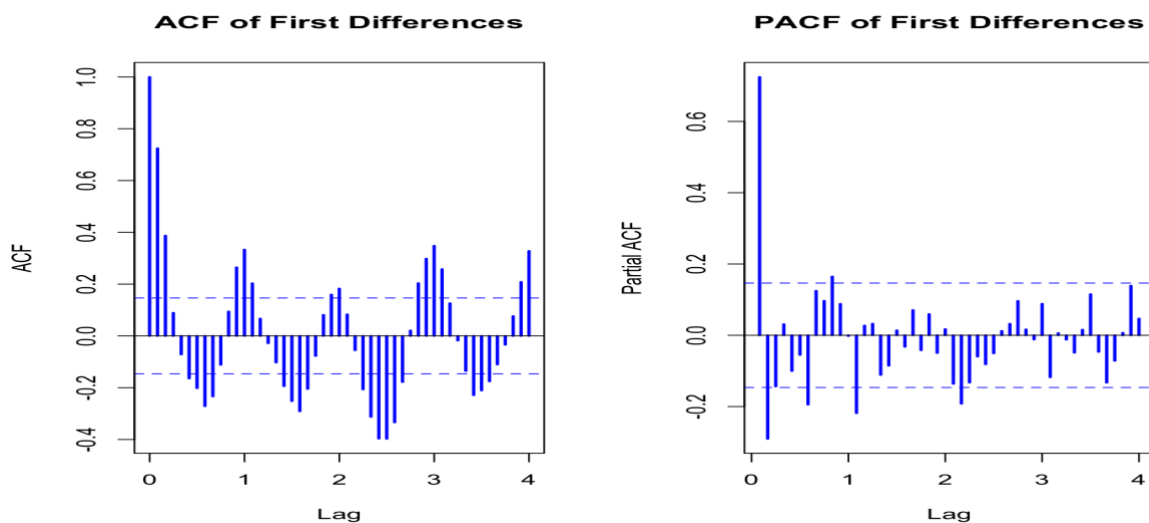


Figure 4. Sample ACF and PACF of the first-differenced chicken series

Figure 4 displays the ACF gradually decays rather than cutting off sharply, while the PACF contains significant spikes at the first few lags. These patterns support considering low-order nonseasonal AR and MA terms but the plot don't identify a unique model.

The plots also contain dependence around the seasonal 12, 24, 36 and 48, indicating that seasonal AR or MA terms with period 12 may be useful. Therefore, several candidate SARIMA models will be fitted and compared using AICc, coefficient significance, and residual diagnostics.

II. Fitting the Model

1. Candidate Models

The `auto.arima()` function was first used to identify a reasonable starting model by minimizing the AICc while restricting the search to models with $d=1$, $D=0$, and no drift.

```
Series: chicken
ARIMA(2,1,0)(2,0,0)[12]

Coefficients:
      ar1      ar2      sar1      sar2
    0.9137 -0.2325  0.2990  0.1176
s.e.  0.0735   0.0739  0.0735  0.0759

sigma^2 = 0.3883: log likelihood = -168.65
AIC=347.3   AICc=347.64   BIC=363.23
```

Figure 5. Output from `auto.arima()` showing the selected starting SARIMA model

The selected model is **SARIMA(2,1,0) × (2,0,0)₁₂** with AIC = 347.30 and AICc = 347.64. It is used as a starting point rather than the final choice. Additional candidate models will be fitted and compared using AIC, AICc, BIC, coefficient significance, and residual diagnostics before selecting the final model.

2. Compare Candidate Models

Three candidate SARIMA models were fitted using maximum likelihood estimation with the forecast: `Arima()` function to ensure that their information criteria were computed consistently.

	Model	AIC	AICc	BIC	Variance
2	SARIMA(2,1,0)(1,0,1)[12]	328.7531	329.0999	344.6900	0.3347
3	SARIMA(1,1,2)(1,0,1)[12]	329.1691	329.6575	348.2934	0.3344
1	SARIMA(2,1,0)(2,0,0)[12]	347.2974	347.6443	363.2344	0.3883

Figure 6. Comparison of candidate SARIMA models

Among the three candidate models, **SARIMA(2,1,0)(1,0,1)₁₂** has the smallest AIC, AICc, and BIC. Its AICc is 329.10, compared with 329.66 for SARIMA(1,1,2)(1,0,1)₁₂ and 347.64 for the automatic starting model. In addition, the two leading models have similar estimated variances of 0.3347 and 0.3344, but the competing model uses one additional ARMA parameter. Therefore, **SARIMA(2,1,0)(1,0,1)₁₂** is selected as the preferred candidate for further evaluation through coefficient significance and residual diagnostics.

3. Selected SARIMA Model

Coefficients:

	Estimate	SE	t.value	p.value
ar1	0.8952	0.0734	12.1975	0.000
ar2	-0.2227	0.0741	-3.0048	0.003
sar1	0.9794	0.0262	37.4394	0.000
sma1	-0.8632	0.0865	-9.9751	0.000

sigma^2 estimated as 0.3271767 on 175 degrees of freedom

AIC = 1.836609 AICc = 1.837894 BIC = 1.925642

Figure 7. Raw sarima() output for the selected SARIMA(2,1,0)(1,0,1)₁₂ model

The selected **SARIMA(2,1,0)(1,0,1)₁₂** model was refitted using sarima() function from the astsa package to obtain coefficient estimates, standard errors, test statistics, p-values, information criteria and diagnostic statistics.

All four estimated coefficients are statistically significant ($p < 0.01$), providing strong support for retaining each parameter in the final model.

4. Estimated Model Equation

Recall that a general SARIMA model has the form:

$$\phi(B)\Phi(B^S)(1-B)^d(1-B^S)^D X_t = \theta(B)\Theta(B^S)W_t \text{ where } (p, d, q) \times (P, D, Q)$$

For the selected model **SARIMA(2,1,0)(1,0,1)₁₂** we have: $p = 2$, $d = 1$, $q = 0$ and $P = 1$, $D = 0$, $Q = 1$, $S = 12$

The model can be written as follows: $\phi(B)\Phi(B^{12})(1-B)X_t = \theta(B)\Theta(B^{12})W_t$

From Figure 7, we have the estimated coefficients: $\hat{\phi}_1 = 0.8952$, $\hat{\phi}_2 = -0.2227$, $\hat{\Phi}_1 = 0.9794$ and $\hat{\Theta}_1 = -0.8632$

Substitute the estimated coefficients gives the fitted equation:

$$(1 - 0.8952B + 0.2227B^2)(1 - 0.9794B^{12})(1 - B)X_t = (1 - 0.8632B^{12})W_t \text{ where } W_t \sim WN(0, 0.3272)$$

5. Diagnostic Analysis

The selected model was fitted using `sarima()`, which automatically produced the standardized residual plot, residual ACF, Normal Q–Q plot, and Ljung–Box p-value plot shown in Figure 8.

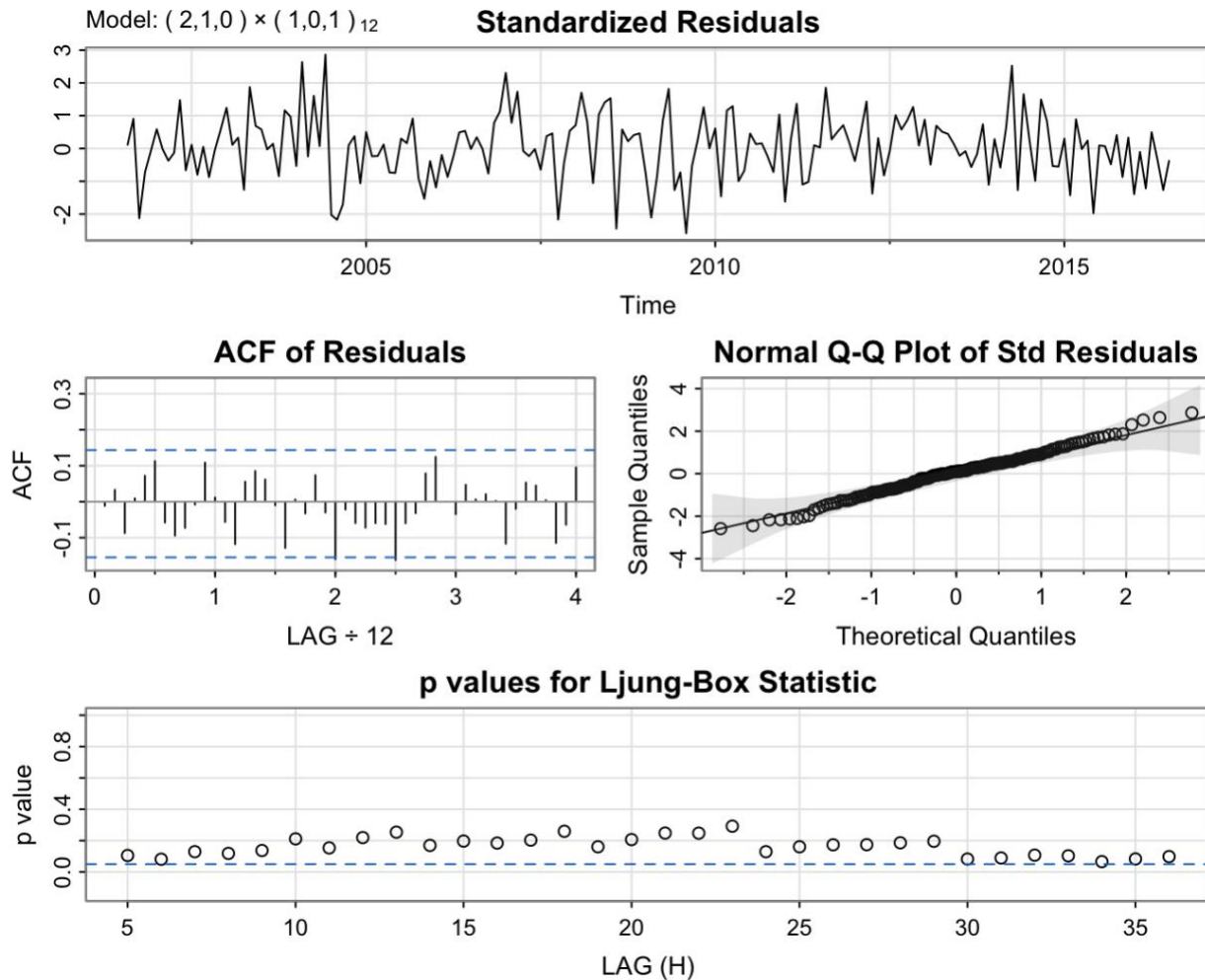


Figure 8. Diagnostic plots for the selected $SARIMA(2,1,0)(1,0,1)_{12}$

Figure 8 shows that the standardized residuals fluctuate randomly around zero with constant variance. The residual ACF contains no significant spikes outside the confidence limits, indicating that little autocorrelation remains. The Q–Q plot is close to a straight line, with only slight departures in the tails, suggesting that the residuals are normally distributed. The Ljung–Box p-values are above 0.05, providing no evidence of remaining serial correlation.

Formal Ljung–Box tests

The Ljung–Box tests produced p-values of 0.219 at lag 12 and 0.129 at lag 24. Since both p-values exceed 5% significance level, the null hypothesis of no residual autocorrelation is not rejected. These results confirm that the residuals behave as white noise and that the selected model provides an adequate fit.

6. Analysis of the Competing Model

```
Coefficients:
      Estimate      SE t.value p.value
ar1    0.4658 0.1250  3.7252  0.0003
ma1    0.4250 0.1297  3.2768  0.0013
ma2    0.2096 0.1063  1.9715  0.0503
sar1   0.9773 0.0277 35.2869  0.0000
sma1  -0.8577 0.0866 -9.9065  0.0000

sigma^2 estimated as 0.3250783 on 174 degrees of freedom

AIC = 1.838933  AICc = 1.840871  BIC = 1.945773
> alternative_lb12

      Box-Ljung test

data:  alternative_residuals
X-squared = 8.1915, df = 7, p-value = 0.316

> alternative_lb24

      Box-Ljung test

data:  alternative_residuals
X-squared = 24.173, df = 19, p-value = 0.1896
```

Figure 9. Raw sarima() output and Ljung–Box tests for the competing SARIMA(2,1,0)(1,0,1)₁₂ model

The principal competing model is SARIMA(1,1,2)(1,0,1)₁₂. Using the coefficients in Figure 9, its fitted equation is:

$$(1 - 0.4658B)(1 - 0.9773B^{12})(1 - B)X_t = (1 + 0.4250B + 0.2096B^2)(1 - 0.8577B^{12})W_t$$

where $W_t \sim WN(0, 0.3251)$

Its Ljung-Box p-values are 0.316 at lag 12 and 0.1896 at lag 24 above the significance level 0.05 meaning the null hypothesis is not rejected, so its residual diagnostics are acceptable. However, its AICc of 329.66 is 0.56 larger than the selected model's AICc, it uses one additional ARMA coefficient, and its MA(2) estimate is only marginally significant ($p = 0.0503$). The selected **SARIMA(2,1,0)(1,0,1)₁₂** model is therefore preferred on information criteria and parsimony.

III. Forecasting with the Selected Model

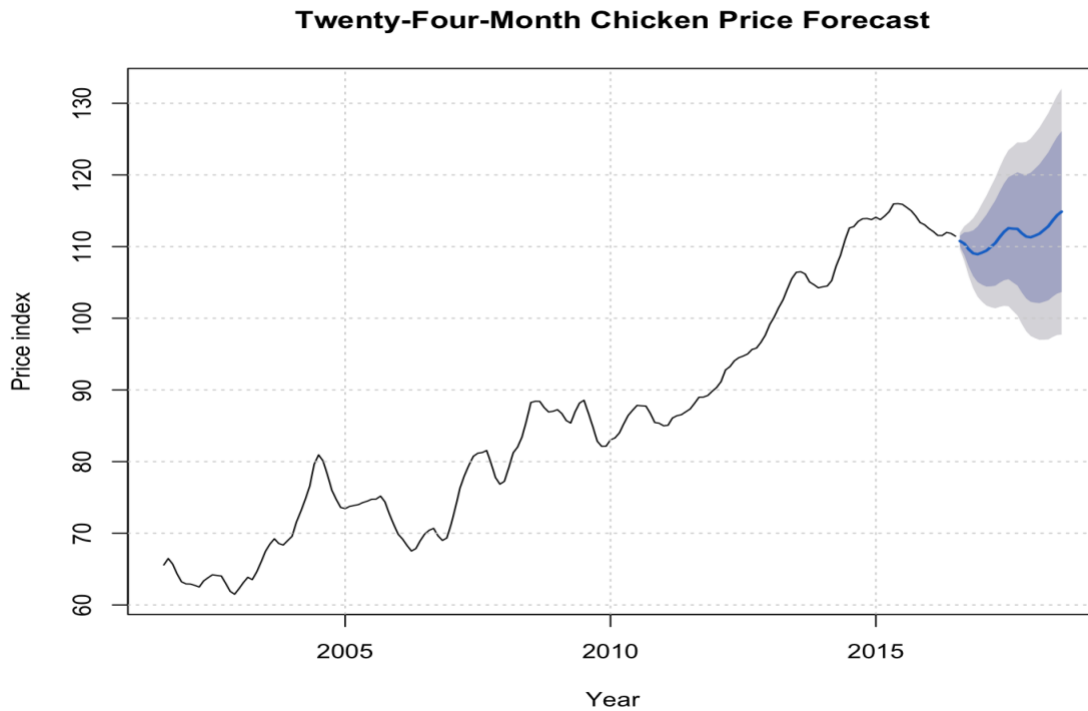


Figure 10. Twenty-four-month forecasts and 80% and 95% prediction intervals from the selected model

After comparing the candidate models, the selected $\text{SARIMA}(2,1,0)(1,0,1)_{12}$ model was refitted using `forecast::Arima()` and used to generate 24 monthly forecasts, covering two complete annual cycles beyond July 2016.

The point forecast initially decreases from 110.7824 in August 2016 to 108.9355 in December 2016. Then, it recovers and retains a modest annual pattern generated by the seasonal AR and MA components. By July 2018, the point forecast reaches approximately 114.8790.

The 95% prediction interval is [109.6473, 111.9175] for August 2016 compared with [97.7238, 132.0361] for July 2018. So, the prediction intervals increase in width as the forecasting horizon increases.

Since the fitted model contains one ordinary difference and no drift, its long-run forecast is driven by the estimated nonseasonal and seasonal dynamics rather than by a deterministic linear trend. So, accumulation of uncertainty in the differenced process also explains the widening prediction intervals.

IV. Conclusion

The original chicken series was nonstationary and required one ordinary difference but no seasonal difference. The ACF and PACF of the first differences supported low-order nonseasonal terms and seasonal dependence at period 12.

The automatic search recommended a $SARIMA(2,1,0)(2,0,0)_{12}$ model as a starting point while comparison of three fitted candidates favored $SARIMA(2,1,0)(1,0,1)_{12}$.

All four coefficients of the selected model were statistically significant, and residual diagnostics indicated that the remaining residuals behaved as white noise. The Ljung–Box tests found no evidence of significant residual autocorrelation, confirming that the model adequately captures both the short-run and seasonal dependence in the data.

The selected model was used to generate 24-month forecasts from August 2016 through July 2018. The point forecasts initially decrease from 110.78 in August 2016 to 108.94 in December 2016, before recovering and reaching 114.88 in July 2018. The seasonal AR and MA components produce a modest repeating annual pattern in the forecast. The 95% prediction intervals increase in width as the forecasting horizon increases.

Overall, the selected $SARIMA(2,1,0)(1,0,1)_{12}$ model provides a good balance of fit, parsimony, and forecasting performance for the chicken price-index series.

Appendix A: Complete R Code

TIME SERIES ANALYSIS OF THE CHICKEN DATASET

1. Load the packages & dataset

```
# Import the library
library(astsa)
library(forecast)
library(fUnitRoots)
```

```
data(chicken)
```

```
# Explore the dataset
head(chicken, 10)
start(chicken)
end(chicken)
frequency(chicken)
length(chicken)
```

2. Preliminary Analysis

Generates Figure 1: Plot the original chicken series

```
plot(chicken, type = "l", lwd = 2, col = "blue", xlab = "Year", ylab = "Price index",
     main = "Monthly U.S. Chicken Price Index")
grid()
```

Generates Figure 2: Month plot used to check for seasonality

```
monthplot(chicken, col.base = "blue", col.mean = "orange", lwd = 1.5, xlab = "Month",
          ylab = "Price index", main = "Seasonality Plot")
```

Apply one ordinary difference

```
dchicken <- diff(chicken)
```

Generates Figure 3: Plot first-differenced chicken series

```
plot(dchicken, type = "h", lwd = 1.3, col = "blue", xlab = "Year", ylab = expression(Delta * X[t]),
     main = "First Differences of the Chicken Series")
abline(h = 0, lty = 2, col="gray40")
grid()
```

Calculate the sample mean of the first differences

```
mean(dchicken)
```

Apply the ADF tests for the original and first-differenced series

```
adf_original <- adfTest(chicken, lags = 5, type = "c")
adf_differenced <- adfTest(dchicken, lags = 5, type = "c")
adf_original
adf_differenced
```

Generates Figure 4: ACF and PACF of the first differences

```
par(mfrow = c(1, 2))
acf(dchicken, lag.max = 48, lwd = 2, col = "blue", main = "ACF of First Differences")
pacf(dchicken, lag.max = 48, lwd = 2, col = "blue", main = "PACF of First Differences")
par(mfrow = c(1, 1))
```

3. Automatic starting model

Using auto.arima as starting model for model identification

```
auto_fit <- auto.arima(chicken, d = 1, D = 0, seasonal = TRUE, ic = "aicc",
                      trace = TRUE, approximation = FALSE, allowdrift = FALSE, allowmean = FALSE)
```

Generates Figure 5: Display the selected model

```
auto_fit
```

4. Fit and compare candidate models

Candidate 1: SARIMA(2,1,0)(2,0,0)[12]

```
fit1 <- Arima(chicken, order = c(2, 1, 0), seasonal = list(order = c(2, 0, 0), period = 12),
              include.drift = FALSE, method = "ML")
```

Candidate 2: SARIMA(2,1,0)(1,0,1)[12]

```
fit2 <- Arima(chicken, order = c(2, 1, 0), seasonal = list(order = c(1, 0, 1), period = 12),
              include.drift = FALSE, method = "ML")
```

Candidate 3: SARIMA(1,1,2)(1,0,1)[12]

```
fit3 <- Arima(chicken, order = c(1, 1, 2), seasonal = list(order = c(1, 0, 1), period = 12),
              include.drift = FALSE, method = "ML")
```

Compare candidates using AICc

```
comparison <- data.frame(
  Model = c("SARIMA(2,1,0)(2,0,0)[12]",
            "SARIMA(2,1,0)(1,0,1)[12]",
            "SARIMA(1,1,2)(1,0,1)[12]"),
  AIC = c(AIC(fit1), AIC(fit2), AIC(fit3)),
  AICc = c(fit1$aicc, fit2$aicc, fit3$aicc),
  BIC = c(BIC(fit1), BIC(fit2), BIC(fit3)),
  Variance = c(fit1$sigma2, fit2$sigma2, fit3$sigma2))
```

Generates Figure 6: Sort by AICc and round the numerical results

```
comparison <- comparison[order(comparison$AICc), ]
comparison[, c("AIC", "AICc", "BIC", "Variance")] <- round(comparison[, c("AIC", "AICc", "BIC", "Variance")], digits = 4)
comparison
```

5. Selected SARIMA model & Diagnostics

```
# Fit the selected model SARIMA(2,1,0)(1,0,1)[12]
# Generates Figure 7 (model output) and Figure 8 (diagnostic plots)
selected_sarima <- sarima(chicken,2, 1, 0, 1, 0, 1, 12,no.constant = TRUE)

# Ljung-Box test through one and two seasonal cycles.
# Four fitted ARMA coefficients imply fitdf = 4.
selected_residuals <- residuals(selected_sarima$fit)
lb12 <- Box.test(selected_residuals, lag = 12, type = "Ljung-Box",fitdf = 4)
lb24 <- Box.test(selected_residuals, lag = 24, type = "Ljung-Box",fitdf = 4)

# Print the complete test results
lb12
lb24
```

6. Analysis of the Competing model

```
# Generates Figure 9: Competing model SARIMA(1,1,2)(1,0,1)[12]
alternative_sarima <- sarima(chicken,1, 1, 2, 1, 0, 1, 12,no.constant = TRUE)
alternative_residuals <- residuals(alternative_sarima$fit)
alternative_lb12 <- Box.test(alternative_residuals,lag = 12, type = "Ljung-Box",fitdf = 5)
alternative_lb24 <- Box.test(alternative_residuals,lag = 24,type = "Ljung-Box",fitdf = 5)

alternative_lb12
alternative_lb24
```

7. Forecasting

```
# Refit the selected model as a forecast::Arima object
final_fit <- Arima(chicken,order = c(2, 1, 0), seasonal = list(order = c(1, 0, 1), period = 12),
  include.drift = FALSE, method = "ML")
# Display the fitted model
final_fit

# Generate 24 monthly forecasts with 80% and 95% intervals
final_forecast <- forecast(final_fit, h = 24, level = c(80, 95))
# Print the complete forecast table
final_forecast

# Generates Figure 10: Plot the 24-month forecast
plot(final_forecast,xlab = "Year", ylab = "Price index",
  main = "Twenty-Four-Month Chicken Price Forecast")
grid()
```

Disclosures:

- (a) I completed this assignment on my own and did not collaborate with any classmates.
- (b) Yes, I used ChatGPT to improve my R code.
- (c) I have read and understood the assignment policies.